

Progress Test 3.B — Elec-S1
3.B · Thévenin’s and Norton’s equivalence theorems
Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. From Norton to Thévenin (1 mark)

A Norton model has $I_{No} = 2.0\text{mA}$ and $R_{No} = 4.7\text{k}\Omega$. Give the electromotive force E_{Th} of the equivalent Thévenin model.

$E_{Th} = R I_{No} = 4.7 \times 10^3 \Omega \times 2.0 \times 10^{-3} \text{ A}$	$E_{Th} = 9.4 \text{ V}$	(with
$R_{Th} = R_{No} = 4.7 \text{ k}\Omega$		

Marking

①	
0	1
☐	☐

Marking scheme
① 1 pt $E_{Th} = 9.4 \text{ V}$ ($\text{k}\Omega \times \text{mA} = \text{V}$).

Exercise 2. Thévenin's theorem

(5 marks)

State Thévenin's theorem for a network made of ideal sources and ohmic conductors, seen from its terminals A and B (draw the model). Explain how E_{Th} and R_{Th} are determined.

Seen from A and B , the network is equivalent to a single ideal voltage source E_{Th} in series with a single resistor R_{Th} :	
$u = E_{Th} - R_{Th} i$ (generator convention).	
E_{Th} = open-circuit voltage between A and B ($i = 0$), with the sources left active (use the divider, Kirchhoff, ...).	
R_{Th} = resistance seen between A and B with all sources switched off : ideal voltage sources replaced by short circuits, ideal current sources by open circuits.	

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Marking scheme

- ① **2 pts** Model drawn: E_{Th} in series with R_{Th} (1 pt); $u = E_{Th} - R_{Th} i$ or an equivalent statement (1 pt).
 ② **2 pts** E_{Th} = open-circuit voltage (1 pt), sources active and $i = 0$ (1 pt).
 ③ **1 pt** R_{Th} with the sources off: voltage sources shorted, current sources opened.

Exercise 3. Thévenin and Norton models

(3 marks)

To switch off an ideal voltage source, it is replaced by:

- ☐ an open circuit ☒ a short circuit (a wire)
☐ a resistor R_{Th} ☐ an ideal current source

Why: a switched-off voltage source imposes $u = 0$ whatever the current: that is a wire. A switched-off current source imposes $i = 0$: that is an open circuit.

A Thévenin model has $E_{Th} = 12\text{ V}$ and $R_{Th} = 3.0\ \Omega$. The equivalent Norton model has:

- ☐ $I_{No} = 36\text{ A}$, $R_{No} = 3.0\ \Omega$ ☒ $I_{No} = 4.0\text{ A}$, $R_{No} = 3.0\ \Omega$
☐ $I_{No} = 4.0\text{ A}$, $R_{No} = 0.33\ \Omega$ ☐ $I_{No} = 0.25\text{ A}$, $R_{No} = 3.0\ \Omega$

Why: $I_{No} = E_{Th}/R = 12/3.0 = 4.0\text{ A}$, and the resistance is common to both models.

A Thévenin (or Norton) model can describe a “black box” containing:

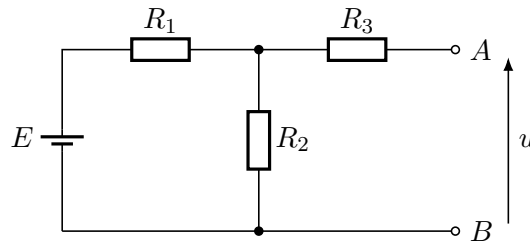
- ☐ a diode;
☒ only ideal sources and ohmic conductors;
☐ a lamp;
☐ a motor.

Why: the models are valid only for ideal sources and ohmic conductors: “no lamp, no motor, no diode” (notes, 3.B).

Exercise 4. Thévenin model of a network

(5 marks)

Determine the Thévenin model (E_{Th} , R_{Th}) of the network below, seen from the terminals A and B . Deduce the current I_{No} of the equivalent Norton model.



$$E = 12 \text{ V}$$

$$R_1 = 6.0 \text{ k}\Omega$$

$$R_2 = 3.0 \text{ k}\Omega$$

$$R_3 = 1.0 \text{ k}\Omega$$

Sources off (E replaced by a wire): R_1 and R_2 are then in parallel, and this pair is in series with R_3 :

$$R_{Th} = R_3 + \frac{R_1 R_2}{R_1 + R_2} = 1.0 + \frac{18}{9.0} = 1.0 + 2.0 \quad \boxed{R_{Th} = 3.0 \text{ k}\Omega}$$

Open circuit ($i = 0$): no current in R_3 , so no voltage across it and u_{AB} equals the voltage across R_2 (voltage divider):

$$E_{Th} = \frac{R_2}{R_1 + R_2} E = \frac{3.0}{9.0} \times 12 \text{ V} \quad \boxed{E_{Th} = 4.0 \text{ V}}$$

$$I_{No} = \frac{E_{Th}}{R_{Th}} = \frac{4.0 \text{ V}}{3.0 \text{ k}\Omega} \quad \boxed{I_{No} = 1.3 \text{ mA}} \quad (4/3 \text{ mA})$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt Sources off: E short-circuited, $R_{Th} = R_3 + R_1 \parallel R_2$.
- ② 1 pt $R_{Th} = 3.0 \text{ k}\Omega$.
- ③ 1 pt Open circuit: no current in R_3 , so a voltage divider across R_2 .
- ④ 1 pt $E_{Th} = 4.0 \text{ V}$.
- ⑤ 1 pt $I_{No} = E_{Th}/R_{Th} = 1.3 \text{ mA}$.

Examiner's tip

R_3 changes R_{Th} but not E_{Th} : in open circuit it carries no current. Check I_{No} directly: with A and B short-circuited, $R_2 \parallel R_3 = 0.75 \text{ k}\Omega$, so E delivers $12/6.75 = 16/9 \text{ mA}$, of which R_3 takes $\frac{R_2}{R_2 + R_3} = \frac{3}{4}$: $I_{No} = \frac{3}{4} \times \frac{16}{9} = \frac{4}{3} \text{ mA}$. ✓

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20